Victorian Detective Fiction and the Nature of Evidence

The Scientific Investigations of Poe, Dickens, and Doyle

Lawrence Frank

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Part One
Edgar Allan Poe

Nebula in Orion, from John Pringle Nichol's *Views of the Architecture of the Heavens*, 3rd edn (1839). Image reproduced courtesy of the History of Science Collections, University of Oklahoma.

1

"The Murders in the Rue Morgue": Edgar Allan Poe's Evolutionary Reverie

> Origin of man now proved. – Metaphysic must flourish. – He who understands baboon ⟨will⟩ would do more towards metaphysics than Locke
>
> Charles Darwin, Notebook M, 16 August 1838

In *The Dialogic Imagination* Mikhail Bakhtin uses evolutionary metaphors to argue that as historical circumstances change, certain literary genres become extinct or endure at best as living fossils, while new genres appear in response to new and perplexing situations. Of the novel, Bakhtin observes:

> Among genres long since completed and in part already dead, the novel is the only developing genre. It is the only genre that was born and nourished in a new era of world history and therefore it is deeply akin to that era, whereas the other major genres entered that era as already fixed forms, as an inheritance, and only now are they adapting themselves – some better, some worse – to the new conditions of their existence.[1]

Such new eras of history appear perhaps after a period of transition, even of crisis, that precedes the fragile consolidation of what may only seem to be a newly-triumphant and prevailing worldview. From such a perspective, Michael McKeon has suggested that changes in genre and in the generic categories of the literary historian register "an epistemological crisis, a major cultural transition in attitudes toward how to tell the truth in narrative."[2] Bakhtin and McKeon imply that both generic classes and truth become relative, products of a particular historical situation. New genres not only posit new truths; they provide narrative

models for arriving at the truth of the moment. It is possible, then, to argue that the detective story, not unlike the novel, appeared at a moment of social and intellectual crisis that it both recognized and defined, even as it promoted the transition from one worldview, generally Christian and ahistorical, to another, profoundly historical worldview that remains to this day unsettling.[3]

A case in point is Edgar Allan Poe's "The Murders in the Rue Morgue" (1841), arguably the first detective story, the first indication of a new genre "born and nourished in a new era of world history." The story introduces the eccentric detective in the character of C. Auguste Dupin as he solves the grotesque murders of Madame and Mademoiselle L'Espanaye (mother and daughter) who were brutally killed and mutilated in their locked apartment in the Rue Morgue by an unknown intruder that will prove to be an Ourang-Outang transported to Paris from the Malay archipelago. Through allusions to the nebular hypothesis of Pierre Simon Laplace (1749–1827) "The Murders in the Rue Morgue" hinted at the existence of a contingent universe denied the trappings of divinity in which the detective finds himself as he sets out to render the apparently unintelligible intelligible without recourse to supernatural explanations. The story appeared in 1841 at a time in which both a resurgent evangelicalism and a conservative Natural Theology were confronted by a positivist science that was to have its nineteenth-century culmination in Charles Darwin's *Origin of Species*.[4] Of course, transitions in worldview do not occur within a few decades; no worldview ever sweeps the field. The mysterious universe of *The Pilgrim's Progress* (1678), in which divine intervention and the supernatural abound, was never to be displaced completely by that version of what was to become Natural Theology that Robinson Crusoe enacts in Daniel Defoe's 1719 novel. The transition from the Natural Theology of William Paley (1743–1805) or of the authors of the Bridgewater Treatises, with their attempts to reconcile science and revealed religion, to a Darwinian universe of natural law and chance has never been consolidated within Western consciousness – as the rise of scientific creationism and, more recently, of the argument for intelligent design reveals.[5]

As a representative of detective fiction, "The Murders in the Rue Morgue" participates in an endless conversation, perhaps a cacophony of dispute, that has never ended.[6] The story itself depicts a culture in which disagreement and discord are everywhere. Men and women of different nationalities and languages are gathered in Paris, a world-city

that becomes a nineteenth-century Babel. The witnesses to the carnage in the house in the Rue Morgue are unable to agree about the language spoken in the shrill – or was it a harsh? – voice that they had heard, about the gender of the speaker, or perhaps about anything else. They speak, literally and figuratively, in foreign languages to each other. The situation points to the debates of the 1830s and 1840s about the origins and meaning of the universe. When C. Auguste Dupin dismisses the acumen of the Paris police as mere cunning, he observes that "there is no method in their proceedings, beyond the method of the moment."[7] With these words Dupin is not simply endorsing a generalized scientific and empiricist method to be demonstrated through his solution to the murders that occurred within the locked apartment of the L'Espanayes. Rather, he alludes to a specific worldview and to the controversies surrounding Pierre Simon Laplace's nebular hypothesis and to one version of it set forth by John Pringle Nichol (1804–1859), Regius professor of astronomy at the University of Glasgow. In his scorn for the method, the common sense, of the moment, Dupin will ally himself with a vision of the universe that, as its critics understood, could lead to religious skepticism.[8]

All of this is clarified in the episode in which Dupin engages in what will prove to be his first tour de force of detection for our benefit. Dupin and the narrator have been walking at night through the streets of Paris without having spoken for some fifteen minutes. Suddenly, Dupin remarks to his companion: "He is a very little fellow, that's true, and would do better for the *Théâtre des Variétés*" ("Murders," p. 402). The astounded narrator feels that Dupin has fathomed his soul; he demands to know the method – "if method there is" – that according to Dupin has led to the links in a chain that run "Chantilly [the cobbler turned tragedian], Orion, Dr. Nichol, Epicurus, Stereotomy, the street stones, the fruiterer" (p. 403). In his explanation of his inferences Dupin will demonstrate a method and, in discussing the topics the narrator dwells upon, allude to an embryonic worldview. Poe's narrator may emphasize the almost preternatural nature of Dupin's gifts; but Dupin himself scorns recourse to the supernatural: he pursues explanations always grounded in empiricist assumptions about causation and human consciousness. In retracing his companion's thoughts, Dupin uses the reconstructive models of nineteenth-century historical disciplines associated with geology and archaeology and later with evolutionary biology. He returns to a posited "starting-point" – the narrator's encounter with a fruiterer that had "thrust [him] upon a pile of paving-

stones collected at a spot where the causeway is undergoing repair" (p. 403). From this accidental event Dupin creates a chronological, an associative, and, implicitly, a causal sequence of ideas based on the assumption that law governs not only all natural phenomena, but also the workings of the human mind.[9]

From the start, the episode touches upon debates central to a time when nineteenth-century geology, particularly that of Charles Lyell and his *Principles of Geology* (1830–33), challenged the literal status of Genesis and a chronology claiming that the universe had been created in 4004 B.C. The narrator's meditations have begun after the brush with the fruiterer. Dupin observes: "You stepped upon one of the loose fragments [of paving-stone], slipped, slightly strained your ankle, appeared vexed or sulky, muttered a few words, turned to look at the pile, and then proceeded in silence" ("Murders," p. 403). The loss of his footing initiates the narrator's musings upon origins. With challenges to Genesis and to traditional conceptions of time, the ground beneath his feet has shifted; he undergoes a version of what Darwin experienced on the voyage of the *Beagle* during the Chilean earthquake of 1835: "A bad earthquake at once destroys the oldest associations: the world, the very emblem of all that is solid, has moved beneath our feet like a crust over a fluid; – one second of time has conveyed to the mind a strange idea of insecurity, which hours of reflection would never have created."[10] When he comes upon what seems to be firmer footing in the interlocked paving-stones, the narrator only finds that the term "Stereotomy" reminds him of the chaos of atoms from which the universe had arisen, by chance, according to the Epicurean cosmogony of Lucretius's *De rerum natura*.

It is at this point that the narrator thinks of Dr. Nichol, the author of *Views of the Architecture of the Heavens* (1837). Except for a reference to the "late nebular cosmogony," Dupin neglects to explain how thoughts of Nichol might lead the narrator to look up to Orion. But anyone turning to Nichol's book would find that it offered an exposition of the nebular hypothesis that Laplace first sketched out in 1796 in his *Exposition du système du monde*.[11] In place of Isaac Newton's world-machine, created and periodically regulated by God, Laplace posited the existence of an original matter existing in a heated, gaseous form that, through the laws of physics, cooled over vast periods of time. As the matter cooled, it began to rotate around emerging centers due to the force of gravity; in this way stars and solar systems would be formed. One such center of cooling gas was to become our sun, orbited by planets that were the condensed remnants of rings of vapor thrown off by

centrifugal force from the cooling, rotating sun. Laplace's hypothesis had been influenced by the astronomer, William Herschel (1738–1822), who had claimed to have found within the constellation Orion a luminous nebula of heated gases. Those gases were the precursors of yet other stars and solar systems, a revelation from the depths of space and time of their origins. It was Herschel who wrote in 1789 of the histories of different celestial systems as akin to "the germination, blooming, foliage, fecundity, fading, withering, and corruption of a plant."[12] Both Herschel and Laplace were to contribute to a worldview that, if Herschel's account is to be trusted, would lead Laplace in his confident theism to observe to Napoleon that he had no need to turn, as Newton had, to a First Cause to explain the origins of the universe.[13]

With its materialist implications, the nebular hypothesis could be associated, as indeed it was, with a Lucretian atheism, a sweeping away of the foundations of everyday life. Early in the nineteenth century William Paley attacked the nebular hypothesis, citing it as one among several hypotheses about the origins of the universe offered by "those who reject an intelligent Creator."[14] Later in the century, in the first of the Bridgewater Treatises written to promote Natural Theology, Thomas Chalmers (1780–1847) specifically mentioned Laplace, holding him responsible for "the tendency of atheistical writers . . . to reason exclusively on the laws of matter, and to overlook its dispositions": "Could all the beauties and benefits of the astronomical system be referred to the single law of gravitation, it would greatly reduce the strength of the argument for a designing cause. . . . as if to fortify still more the atheism of such a speculation[,] . . . La Place [*sic*] would have accredited the law, *the unconscious and unintelligent law, that thing according to him of blind necessity*, with the whole of this noble and beautiful result," that is the solar system and the universe.[15] Writing in 1833, Chalmers sensed to what speculations the nebular hypothesis might lead: "Insomuch, that though we conceded to the atheist, the eternity of matter, and the essentially inherent character of all its laws – we could still point out to him, in the manifold adjustments of matter . . . the most impressive signatures of a Deity" (pp. 19–20): "That *blind and unconscious matter* cannot, by any of *her* combinations, evolve the phenomena of mind, is a proposition seen in its own immediate light, and felt to be true with all the speed and certainty of an axiom" (p. 27, emphasis added). Of course, not all proponents of the nebular hypothesis perceived themselves either as skeptics or as potential transmutationists. In other Bridgewater Treatises, religious men of science like William Whewell (1794–1866) of Cambridge and William Buckland (1784–1856) of Oxford could

provide thorough expositions of Laplace's ideas: it was, in fact, Whewell who coined the term, nebular hypothesis.[16] Gideon Mantell, the popularizer of Charles Lyell's dynamic geology, could endorse the nebular hypothesis in his *Wonders of Geology* in 1838 and remain thoroughly conventional in his religious beliefs.[17]

There are, then, good reasons for not explaining why the narrator's thoughts have moved from Epicurus to Dr. Nichol, rather than to Whewell, Buckland, or Mantell. Dupin's companion is preoccupied with one version of the nebular hypothesis that can be seen as a challenge to orthodoxies of all sorts – religious, scientific, and social. Nichol was an advocate of political economy and, in its name, social change, one of a group of self-styled reformers including John Stuart Mill (1806–1873) and George Combe (1788–1858), the phrenologist. They invoked the rule of natural law and the notion of a changing universe to attack entrenched social and religious institutions. Inevitably, they found themselves in conflict with defenders of the existing social order – with those like Whewell and Adam Sedgwick (1785–1873), first professor of geology at Cambridge and a patron of Charles Darwin, who tried to dominate the emerging sciences in the name of Anglicanism. In their defense of society as it existed, the dons of Cambridge and Oxford could be expected to respond to Nichol and to others with charges of political radicalism and religious impiety.[18]

In his *Views of the Architecture of the Heavens*, Nichol wrote with a keen awareness of potential critics. Throughout his book Nichol celebrated "that BEING, who amid change alone is unchangeable – whose glance reaches from the beginning to the end – and whose presence occupies all things" to quell "uneasy feelings" aroused by "the idea of a process which may appear to substitute *progress* for *creation*, and place *law* in the room of *providence*."[19] He proceeded to expatiate upon "Laplace's bold and brilliant *induction*" (p. 185) that had revealed a "System of Evolution of which Creation as it exists is only one phase," by which "to develope all possible variety, to exhibit how, without infraction of stedfast law, Being may be infinitely diversified, and room found for unfolding the whole riches of the Almighty" (pp. 45–46): "Who can ascend so far up that vast chain which unites the eternal past with the fleeting present; who – to go no higher – can dwell on the idea of our Sun being born from one of those dim nebulæ, order growing within him by effect of law, and the worlds he illumines and sustains, springing gradually into being, – without engrossing emotions!" (pp. 187–88). Yet such emotions must encompass the possibility that matter itself "may bear within it, laid up in its dark bosom – the germs, the producing powers

of that LIFE, which in coming ages will bud and blossom, and effloresce, into manifold and growing forms" (p. 138). Within a universe so conceived "the existence of the human race is an invisible speck, [that] will have resulted during a stage of condensation in a secondary nebula": "In relation to the nebulæ, Man is only an Ephemeron" (pp. 151–52). Nichol's vision of Man as a "transient organization, with whose progress the education of a Spiritual Being has been for a moment connected" (p. 188) might arouse anxiety, as Nichol substituted for a single act of creation an evolving process governed only by natural law. Yet he argued that such anxiety arises from a misunderstanding, for law cannot be separated from Mind:

> Law is a mere name for a long order – an order unoriginated, unupheld, unsubstantial, whose floor sounds hollow beneath the tread, and whose spaces are all void; an order hanging tremblingly over nothingness, and of which every constituent – every thing and creature fails not to beseech incessantly for a substance and substratum in the idea of ONE – WHO LIVETH FOR EVER!
>
> (pp. 189–90)

Nichol captured the agony of those who feel that they tread upon nothing, only to console his readers with, in the eyes of his critics, a convenient abstraction. He could only further stir uneasiness as he observed that "absolute permanence is visible nowhere around us, and the fact of change merely intimates, that in the exhaustless womb of the future, unevolved wonders are in store." But such wonders require "the close of one mighty cycle in the history of the solar orb – the passing away of arrangements which have fulfilled their objects that they might be transformed into new. Thus is the periodic death of a plant perhaps the essential to its prolonged life, and when the individual dies and disappears, fresh and vigorous forms spring from the elements which composed it" (*Architecture*, pp. 194–95).

It is not surprising that Poe's narrator may not find solace in Nichol's claim that "a time may come, when the veil can be drawn aside – when spirit shall converse *directly* with spirit, and the creature gaze without hinderance on the effulgent face of its Creator" (*Architecture*, p. 196). Only as he finds surer footing in the pavement of the alley, Lamartine, does he regain what Nichol called "the solid ground of fact and observation" (p. 122): at that moment he casts his "eyes upward to the great *nebula* in Orion" ("Murders," p. 404). It was, after all, William Herschel's speculations on the luminous clouds he claimed to have discovered in

the constellation of Orion that had influenced Laplace's formulation of the nebular hypothesis.[20] The narrator gazes upward to glimpse the birth of other stars, perhaps of other solar systems, that could eventually give rise to life itself. Yet, in contemplating Orion, the narrator inevitably is reminded of a recent attack upon the would-be tragedian, Chantilly, in which the critic in the *Musèe* mocked the actor's change of name by citing a Latin line about which he and Dupin have often talked: "Perdidit antiquum litera prima sonum" (p. 404). The line is from the *Fasti*, Ovid's uncompleted calendar that included an account of the birth of Orion. In the *Fasti* Ovid told the story of Jupiter, Neptune, and Mercury, who urinate upon a bull's hide from which, after being covered by earth, a boy is born some ten months later. The child is originally called Urion, from the Greek "ouron," for urine; the name is altered to obscure his true origins and the curious fact that he is not born of woman.[21] The "pungencies connected with this explanation," so Dupin claims, has led the narrator "to combine the two ideas of Orion and Chantilly" (p. 404). In thinking of the cobbler who has changed his name, the narrator thinks of an act of self-transformation from some ignominious condition. Yet the creation of Orion out of urine, earth, and the bull's hide – out of matter in its least pleasant forms – suggests not only the diminutive, perhaps comic Chantilly; it stands as a curious parody of any evolutionary account of human origins that might suggest, as the *Architecture of the Heavens* had, that matter itself bears within it germs that may produce life without an act of special creation. In drawing himself to his full height, Dupin's companion not only thinks of the cobbler's stature; he asserts his own dignity and humanity within the context of certain accounts of human origins. In his erect posture the narrator, who to that point has "been stooping in [his] gait" (p. 404), would seem to shrug off any hypothesis that would suggest that he is related to creatures of some lower order.

Although Dupin's reconstruction of his friend's thoughts would seem to bring to an end the preoccupations that bother him, "The Murders in the Rue Morgue" becomes an extension of the narrator's reverie, even a confirmation of his fears. Dupin's revelation that an Ourang-Outang had been the killer of Madame and Mademoiselle L'Espanaye only further develops the possibilities that have proven to be so disconcerting to the narrator. In his efforts to persuade his friend that the marks on the throat of Mademoiselle L'Espanaye have not been left by a human hand, Dupin turns to a passage from Georges Cuvier: "It was a minute anatomical and generally descriptive account of the large fulvous Ourang-Outang of the East Indian Islands. The gigantic stature,

the prodigious strength and activity, the wild ferocity, and the imitative propensities of these mammalia are sufficiently well known to all. I understood the full horrors of the murder at once" ("Murders," p. 424). Burton R. Pollin has already recognized that Dupin has referred his friend to Cuvier's *Animal Kingdom* (1817), and Pollin has further noted that the entry on the Ourang-Outang does not, in fact, mention the ferocity of the animal.[22] More suggestive is the fact that Cuvier is careful to qualify the claims of those who would argue that the Ourang-Outang approaches "most nearly to Man in the form of his head, height of forehead, and volume of brain." Yet, in a passage omitted from "The Murders in the Rue Morgue" – paralleling the omission of the significance of Dr. Nichol – Cuvier went on to discuss the Ourang-Outang's imitative talents and then turned to deal with the Pongo, which resembles the Ourang-Outang and "in size is nearly equal to Man." The full implications of the entry are revealed in a footnote explaining the etymology of the animal's name: "*Orang* is a Malay word, signifying *reasonable being*, which is applied to man, the ourang-outang, and the elephant. *Outang* means *wild*, or *of the woods*; hence, Wild Man of the Woods."[23] Cuvier's etymology may well be a pure fiction, a repetition in the story of Ovid's account of the derivation of the name, Orion, from Urion. But just as the *Fasti* suggests Orion's humble origins, Cuvier's discussion of the name, Ourang-Outang, connects the animal with the human species, perhaps as its ancestor.

Given the scientific debates of the time and an anxiety about the implications of the nebular hypothesis, someone reading the *Animal Kingdom*, with Cuvier's account of the successive appearance of living organisms over time, might well think of James Burnett, Lord Monboddo (1714–1799), the eighteenth-century Scottish polymath who had become notorious for his account of the origin of language.[24] In *Of the Origin and Progress of Language* (1773–92) and in *Antient Metaphysics* (1779–99), Monboddo, like Cuvier after him, translated Ourang-Outang as "Wild Man of the Woods." For Monboddo, the Ourang-Outang was a necessary stage in the progression of the human species; indeed, he *is* a man as suggested by the name, "*Man of the Woods*, . . . by which he is called by the people of Africa, where he is to be seen, and who do not appear to have the least doubt that he is a man; which, as they live in the country with him, they should know better than we do."[25] Accounts of the Ourang-Outang provide a description of "man, in his first state":

> for man is said first to creep, that is, to go upon all four. . . . After that, he is erected, and gets the use of an artificial weapon, such as the

> Ourang Outang uses. Next, he invents rude and barbarous cries . . . by which men communicated their sensations, appetites, and desires to one another. And, last of all, they formed ideas, and invented words to express them. . . . But this is a step in the progress towards the civilized life, which the Ourang Outang has not yet made.
>
> (IV, 31–32)

In fact, Monboddo argued that the Ourang-Outang has a fully developed larynx – "the upper part of the larynx, which is called in English *the knot of the throat*, and in French *la glotte*" (IV, 110) – exactly like that of man. As the self-proclaimed promoter of the theism of Plato and Aristotle, Monboddo claimed, however improbably, that

> it was of absolute necessity that, in the progress of the human species, man should at some time or another be such an animal: For, if he was originally a quadruped, as I think I have proved by facts incontestible, with only a natural aptitude, more than any other animal, to walk on two, as Aristotle has said, the first step in his progression was to become a biped.
>
> (IV, 26)

As further proof of his claims Monboddo stated that the Ourang-Outang can copulate with human females and produce offspring. Out of a hodge-podge of philosophy, anecdotal reporting, and flights of fancy, Monboddo depicted the Ourang-Outang as man in his natural state before the invention of the art of language that was necessary for his progression towards full humanity.

It is perhaps with such discussions of the Ourang-Outang in mind that Dupin seizes upon the confusing testimony about the language and even the gender of one of the voices heard behind the locked door to the L'Espanaye apartment. He comments: "Now, how strangely unusual must that voice have really been . . . in whose *tones*, even, denizens of the five great divisions of Europe could recognise nothing familiar!" ("Murders," p. 416). With a characteristic racism, Dupin and his friend descend a hierarchy from the apex constituted by speakers of Indo-European languages to Asiatics and Africans, with their "barbarous tongues," to those excluded from the human community through the unintelligibility of the sounds that they produce. In response to Dupin's observation that the voice was "devoid of all distinct or intelligible syllabification," the narrator at last exclaims that "a madman . . . has done this deed – some raving maniac, escaped from a neighboring *Maison de*

Santé" (p. 423). Asians, Africans, lunatics – and perhaps, by implication, women – become associated with the Ourang-Outang. Without language, whatever creature had uttered such sounds and brutally killed the two women is exiled from the human species and ceases, in its apparent motivelessness, to be a responsible agent.[26]

In emphasizing that "madmen are of some nation" and in dismissing "the blundering idea of *motive*" ("Murders," pp. 423, 421), Dupin introduces the Ourang-Outang as the culprit, reducing the murders to purely contingent events without purpose and meaning. He denies that any significance is to be found in the accidental intrusion of the Ourang-Outang into the L'Espanaye apartment or in its frenzied attack upon the mother and the daughter. Yet the fate of the two women demands some response: they had been so violated that the daughter's tongue had been bitten through, the mother's throat cut – I would suggest at the larynx, Monboddo's knot of the throat – so that her head had been severed from her body.[27] A psychological reading of the story would emphasize the misogyny of the murders: the imitative behavior of the Ourang-Outang and the name "Wild Man of the Woods" – suppressed in the story itself – would transform the creature into the Double of its master, the sailor, acting out his fantasies upon the two women. In destroying the women's organs of speech, the Ourang-Outang blindly denies their status as human beings, confirming the darkest implications of male attitudes toward women.[28]

But such a reading complements the one I am suggesting. In the figure of the Ourang-Outang and its attack upon the organs of speech, "The Murders in the Rue Morgue" points to Lord Monboddo's speculations on the progress of the human species in its development from lower animals. As it thrusts the corpse of Mademoiselle L'Espanaye up the chimney, feet first, the Ourang-Outang enacts male terror of the female body and captures, in another sense, the terror and confusion of those who suddenly find themselves in a contingent universe. In his Bridgewater Treatise Thomas Chalmers had said that "blind and unconscious matter cannot, by any of her combinations, evolve the phenomena of mind." In his discussion of the nebular hypothesis John Pringle Nichol had written of the "womb of the future." Both, as men are wont to do, associate matter and nature with the female, implicitly denying women consciousness and the power of speech. But, in association with a rage directed toward women, the murders reveal an act of retribution directed against a universe that is itself the result of an accidental birth out of matter in a gaseous state; that has produced life in ways not fully understood; and that is, even now, moving toward its inevitable death.

In the midst of debates about the implications of the nebular hypothesis, "The Murders in the Rue Morgue" offered all that the critics of Laplace most feared: a harrowing vision of a Lucretian universe. It was a vision recorded by Charles Darwin in his personal notebooks in the spring and summer of 1838. In terms curiously like those we have seen in Poe's detective story, Darwin wrote,

> —look abroad, study gradation. Study unity of type—Study geographical distribution
>
> study relation of fossil with recent. the fabric falls! But Man——wonderful Man. "divino ore versus cœlum attentus" is an exception.— He is Mammalian. . . . he is not a deity, his end 《under present form》 will come, (or how dredfully we are deceived) then he is no exception. . . .
>
> Let man visit Ourang-outang in domestication, hear expressive whine, see its intelligence when spoken; as if it understood every word said . . . 《let [man] look at savage, roasting his parent, naked, artless, not improving yet improvable》 & then let him dare to boast of his proud preeminence.— 《not understanding language of Fuegian, puts on par with Monkeys》[29]

The fabric that falls for Darwin must be that of Natural Theology, with its emphasis on the doctrine of special creation and the uniqueness of humanity. For Darwin, as for Poe, Man is mammalian, not a deity, perhaps separated from other animals only by a face turned toward the heavens and by the power of speech.

In the decade after Darwin's notebook entry it is understandable, then, that opponents of the nebular hypothesis were not to be reassured by the purportedly theistic vision of the 1844 *Vestiges of the Natural History of Creation* or Poe's 1848 lecture on the Universe and its published version, "Eureka."[30] The anonymous author of *Vestiges* – Mr. Vestiges as he came to be called – was Robert Chambers, coeditor of the *Chambers's Edinburgh Journal*, in which, appropriately, an abridged version of "The Purloined Letter" appeared in 1844. John Pringle Nichol would accuse Mr. Vestiges of plagiarism in his uniting of the nebular hypothesis with Lamarckean premises to argue that a "First Cause" – "a primitive almighty will" – has produced the two laws that govern the universe: "The inorganic has one final comprehensive law, GRAVITATION. The organic, the other great department of mundane things, rests in like manner on one law, and that is, – DEVELOPMENT."[31] In the nebula of

Orion we have glimpsed "the process through which a sun goes between its original condition, as a mass of diffused nebulous matter, and its full-formed state as a compact body" (*Vestiges*, p. 8). With the laws of gravitation and development, there is no need to invoke miracle or the doctrine of special creation. Over the course of time the solar system has been formed; even "the first step in the creation of life upon this planet was *a chemico-electric operation, by which simple germinal vesicles were produced*" (pp. 204–5), initiating a process that led from simple organisms to complex ones that include human beings. God willed into existence a process that permitted Mr. Vestiges to deny the dualism of Mind and Matter and to suggest that "the difference between mind in the lower animals and in man is a difference in degree only" (pp. 335–36). There are even echoes of Lord Monboddo as Mr. Vestiges discussed the larynx, the trachea, and the mouth to argue that language is not a new gift of the Creator but that "man" has possessed a physiological organization that was ready for the production of intelligible sound. For those who might be taken aback by his attacks on divine intervention in human affairs or visions of an anthropomorphic God, Mr. Vestiges offered a vista of unending progress in civilization and the further development of human reason in an unimaginable future.

Those who heard Poe's lecture on the Universe and read his "Eureka: A Prose Poem" in 1848 recognized the unacknowledged influence of *Vestiges*.[32] With explicit references to Epicurus, William Herschel, and Nichol's *Architecture of the Heavens* – and a footnote referring to "The Murders in the Rue Morgue" – Poe proceeded to offer his own interpretation of "*the Material and Spiritual Universe: – of its Essence, its Origin, its Creation, its Present Condition and its Destiny*" ("Eureka," p. 1261). Like Mr. Vestiges, Poe wrote of a divine Unity that, through an act of pure volition, "*now* exists solely in the diffused Matter and Spirit of the Universe; and — [of] the regathering of this diffused Matter and Spirit [that] will be but the re-constitution of the *purely Spiritual* and Individual God" (p. 1357). In place of Mr. Vestiges' laws of Gravitation and Development, Poe introduced the rule of Repulsion and Attraction that governs the inanimate and the animate realms. He offered an idiosyncratic pantheism that some associated with Spinoza to explain the origins of the universe and consciousness and to reveal, through the power of Attraction, an inevitable return to the primal unity from which all has come. Unlike Mr. Vestiges, Poe rejected the possibility that "the Universe has no conceivable end": without an end "Creation would have affected us as an imperfect *plot* in a romance, where the *dénoûment* is awkwardly brought about by interposed incidents external and foreign to the main subject;

instead of springing out of the bosom of the thesis" (p. 1352). Human beings demand plots with teleological implications: "The plots of God are perfect. The Universe is a plot of God" (p. 1342). God's plot demands that the universe of matter and spirit through which divinity has diffused itself will finally, and satisfyingly, be rejoined with the unity from which they have proceeded.

But those who were skeptical of the unorthodox implications of the nebular hypothesis would have nothing to do with "the sublime chronology" (*Vestiges*, p. 22) of a Mr. Vestiges or with Poe's version of the universe as a "plot of God." In his anonymous review of *Vestiges* in the *Edinburgh Review*, Adam Sedgwick could write of the book as "the raving madness of hypothetical extravagance," mention Lucretius and his *De rerum natura*, and yoke Mr. Vestiges with Lord Monboddo, each of whom "brooded over the fantasies of his mind till his dreams became to him as substantial realities."[33] The responses to Poe's lecture and "Eureka" were mixed, but the unsigned review of "Eureka" in the *Literary World* attacked Poe's speculations as "extraordinary nonsense, if not blasphemy; and it may very possibly be *both*" (*Poe Log*, p. 746). The unorthodox speculations in *Vestiges* and "Eureka" were properly seen as threats to traditional Christianity and to an existing social order in Great Britain and the United States. Those who would have known of the feverish dreams of Mr. Vestiges or Poe feared that the act of divine volition that created an evolving universe, with or without an end, might well be but a step away from a purely contingent originating event from which the solar system and life itself were to arise. In the place of a divine chronology or a plot of God we may be left perhaps with a romance, a plot in the mind of C. Auguste Dupin, reconstructed only after a series of coincidences that led, with a seeming inevitability, to the murders in the Rue Morgue.

In 1841 "The Murders in the Rue Morgue" explored the implications of the nebular hypothesis at a moment when the contest between differing worldviews had not been decided and when the future could not be predicted. Poe's story did not simply reinforce a prevailing orthodoxy: at the time, Natural Theology and evangelicalism remained powerful voices in the ongoing, and unending, conversation about the origin and nature of the universe. The story may, in fact, have been in the service of an emerging perspective whose full implications were to be clarified in 1859 with the publication of the *Origin of Species*. Poe himself, as "Eureka" suggests, remained undecided, perhaps divided, about the speculations of John Pringle Nichol or Mr. Vestiges. Instead, "The Murders in the Rue Morgue" suggests the situation of the indi-

vidual within a Parisian Babel of controversy, anticipating in its own outré way the possible ascendancy of one evolutionary vision of the universe. In the figure of C. Auguste Dupin, Poe depicted the individual as a creature of history who must reconstruct the past even as he comes upon the fact of the contingency of certain events, without any reassuring knowledge of the future. It becomes a harrowing vision, centered upon the presence of the Ourang-Outang – the ominous "it" – that in a seemingly motiveless frenzy introduces the terror of a history secularized and devoid of design. In such a world there is no longer the solace provided by claims that "the plots of God are perfect. The Universe is a plot of God." Rather, we are left with the reconstructions of a Dupin and, later, of a William Legrand in "The Gold-Bug" (1843), for whom "*perfection* of plot is really, or practically, unattainable . . . because it is a finite intelligence that constructs" ("Eureka," p. 1342). Such plots, like "an imperfect *plot* in a romance" (p. 1352), become a mark of the detective story as it responds to a new era of world history. Through the creation of plot, both Dupin and Legrand struggle against that "temporary paralysis" that coincidence induces: each establishes "a connexion – a sequence of cause and effect" ("Gold-Bug," p. 511) – in the face of what may prove to be mere chance.[34] Through an almost preternatural will to power, Dupin and Legrand create contexts for the literal and figurative texts before them. In this way they regain their footing and walk again in a world that has once been, but may never again be, the "very emblem of all that is solid."

secrecy and solitude, his moods of depression (which he shared with [Poe's] Dupin) are earmarks of the Romantic rebel then and now. Behind Holmes lurk the figures of nineteenth-century poets, Byron certainly, probably Baudelaire, who translated Poe and pressed Poe's guilty knowledge to new limits" (p. 181). In *Bloodhounds of Heaven*: *The Detective from Godwin to Doyle* (Cambridge, Mass.: Harvard University Press, 1976), Ian Ousby traces the changes in Doyle's depiction of Holmes as he ceases to be associated with Decadence and is transformed into the gentleman and the specialist (pp. 158–64).

49. William Whewell, *History of the Inductive Sciences*, *from the Earliest to the Present Times*, 3 vols (London: John W. Parker, 1837), III, 481.
50. Ross Macdonald, "The Writer as Detective Hero," in *Detective Fiction*, p. 186.
51. For recent discussions of the validity of the concept of culture, see Christopher Herbert, *Culture and Anomie*: *Ethnographic Imagination in the Nineteenth Century* (Chicago: University of Chicago Press, 1991); and Alan Liu, "Local Transcendence: Cultural Criticism, Postmodernism, and the Romanticism of Detail," *Representations*, no. 32 (fall 1990): 75–113.
52. See Brooke, *Science and Religion*: "There was no linear succession of scientific insights, inexorably culminating in Darwin's theory of natural selection" (p. 227). Also, see James A. Secord, "Lifting the Veil," in *Victorian Sensation*: *The Extraordinary Publication*, *Reception, and Secret Authorship of "Vestiges of the Natural History of Creation"* (Chicago: University of Chicago Press, 2000), pp. 515–32.

Chapter 1 "The Murders in the Rue Morgue": Edgar Allan Poe's Evolutionary Reverie

1. M. M. Bakhtin, *The Dialogic Imagination: Four Essays*, ed. Michael Holquist, trans. Caryl Emerson and Michael Holquist, University of Texas Press Slavic Series, no. 1 (Austin: University of Texas Press, 1981), p. 4.
2. Michael McKeon, *The Origins of the English Novel, 1600–1740* (Baltimore: Johns Hopkins University Press, 1987), p. 20.
3. I am using Thomas S. Kuhn, *The Structure of Scientific Revolutions*, 2nd edn (Chicago: University of Chicago Press, 1970), to discuss the implications of detective fiction. For other discussions of detective fiction, see John G. Cawelti, *Adventure, Mystery, and Romance: Formula Stories as Art and Popular Culture* (Chicago: University of Chicago Press, 1976); Stephen Knight, *Form and Ideology in Crime Fiction* (Bloomington: Indiana University Press, 1980); Dennis Porter, *The Pursuit of Crime: Art and Ideology in Detective Fiction* (New Haven: Yale University Press, 1981); D. A. Miller, *The Novel and the Police* (Berkeley: University of California Press, 1988); and Ronald R. Thomas, *Detective Fiction and the Rise of Forensic Science*, Cambridge Studies in Nineteenth-Century Literature and Culture, no. 26 (Cambridge, Eng.: Cambridge University Press, 1999).
4. See Charles Coulston Gillispie, *Genesis and Geology: A Study in the Relations of Scientific Thought, Natural Theology, and Social Opinion in Great Britain, 1790–1850* (Cambridge, Mass.: Harvard University Press, 1951).
5. See Neal C. Gillespie, *Charles Darwin and the Problem of Creation* (Chicago: University of Chicago Press, 1979); and George Levine, *Darwin and the Nov-*

elists: Patterns of Science in Victorian Fiction (Cambridge, Mass.: Harvard University Press, 1988). For a discussion of creationism in the United States, see Ronald L. Numbers, *The Creationists* (New York: Alfred A. Knopf, 1992). Also, see Robert T. Pennock, ed., *Intelligent Design Creationism and its Critics: Philosophical, Theological, and Scientific Perspectives* (Cambridge, Mass.: MIT Press, 2001).

6. See Kenneth Burke's "The Philosophy of Literary Form," in *The Philosophy of Literary Form: Studies in Symbolic Action*, 3rd edn (Berkeley: University of California Press, 1973), pp. 110–11.
7. Edgar Allan Poe, "The Murders in the Rue Morgue," in *Edgar Allan Poe: Poetry and Tales*, ed. Patrick F. Quinn (New York: Library of America, 1984), p. 411. All further references to this story and to other Poe works are from this edition and appear in the text.
8. For a discussion of detective fiction, Poe's detective stories, and Poe's knowledge of Laplace, see John T. Irwin, *The Mystery to a Solution: Poe, Borges, and the Analytic Detective Story* (Baltimore: Johns Hopkins University Press, 1994), pp. 322, 338–39, and 352.
9. For discussions of this episode, see Burton R. Pollin, "Poe's 'Murders in the Rue Morgue': The Ingenious Web Unravelled," in *Studies in the American Renaissance, 1977*, ed. Joel Myerson (Boston: Twayne Publishers, 1978), pp. 235–59; and Irwin, *Mystery to a Solution*, pp. 349–56. Pollin suggests that the narrator's train of thought is empty of meaning; Irwin provides a reading dealing with social issues.
10. Charles Darwin, *Journal of Researches into the Geology and Natural History of the Various Countries Visited by H.M.S. Beagle, Under the Command of Captain Fitzroy, R.N. from 1832 to 1836* (London: Henry Colburn, 1839), p. 369.
11. See P[ierre] S[imon] Laplace, *The System of the World*, trans. J. Pond, 2 vols (London: Richard Phillips, 1809), II, 354–75.
12. Quoted in Stephen G. Brush, "The Nebular Hypothesis and the Evolutionary Worldview," *History of Science*, 25 (1987): 251.
13. See Brush, "The Nebular Hypothesis and the Evolutionary Worldview."
14. William Paley, *Natural Theology: or, Evidences of the Existence and Attributes of the Deity, Collected from the Appearances of Nature*, 3rd edn (London: R. Faulder, 1803), p. 414.
15. Thomas Chalmers, *On the Power, Wisdom, and Goodness of God, as Manifested in the Adaptation of External Nature to the Moral and Intellectual Constitution of Man*, A New Edition (Philadelphia: Carey, Lea and Blanchard, 1836), pp. 17–18, emphasis added. All further references are to this edition and appear in the text.
16. See Simon Schaffer, "The Nebular Hypothesis and the Science of Progress," in *History, Humanity and Evolution: Essays for John C. Greene*, ed. James R. Moore (Cambridge, Eng.: Cambridge University Press, 1989), pp. 131–64.
17. For discussions of these issues, see Gillispie, *Genesis and Geology*; and Jack Morell and Arnold Thackray, *Gentlemen of Science: Early Years of the British Association for the Advancement of Science* (Oxford: Clarendon Press, 1981). For a discussion of the response to the nebular hypothesis in the United States, see Ronald L. Numbers, *Creation by Natural Law: Laplace's Nebular Hypothesis in American Thought* (Seattle: University of Washington Press, 1977).
18. See Schaffer, "Nebular Hypothesis and the Science of Progress."

19. J[ohn] P[ringle] Nichol, *Views of the Architecture of the Heavens: In a Series of Letters to a Lady*, 3rd edn (Edinburgh: William Tait, 1839), p. 189. All further references are to this edition and appear in the text.
20. See Brush, "Nebular Hypothesis and the Evolutionary Worldview," and Numbers, *Creation by Natural Law*, for discussions of how Lord Rosse's resolution of the nebula in Orion into already formed stars was later used by proponents of Natural Theology to refute the nebular hypothesis.
21. See *Ovid's Fasti*, trans. Sir James George Frazer, Loeb Classical Library, no. 253 (New York: G. P. Putnam's Sons, 1931), pp. 296–301, v. 493–544.
22. See Pollin, "The Ingenious Web Unravelled," pp. 253–54. Pollin does not deal with Cuvier's explanation of the etymology of the name, Ourang-Outang. For a nineteenth-century discussion of cosmogony and Cuvier, see "Cuvier's Theory of the Globe," review of *A Discourse on the Revolutions of the Surface of the Globe, and the Changes thereby Produced in the Animal Kingdom*, by Baron G. Cuvier, *Southern Review*, 8 (1831–32): 69–88: the essay attacked "the doctrine of atoms, with all its train of atheistical notions," even as it rejected explanations invoking "the immediate agency of the Creator when events can be accounted for by secondary causes"; it concluded by quoting Cuvier's English editor, who refers to the "fertile womb" of nature.
23. Baron Cuvier, *The Animal Kingdom, Arranged According to its Organization, Serving as a Foundation for the Natural History of Animals, and an Introduction to Comparative Anatomy*, 4 vols (London: G. Henderson, 1834–37), I, 47–48.
24. See Hans Aarsleff, *The Study of Language in England, 1780–1860* (Princeton: Princeton University Press, 1967).
25. James Burnett, Lord Monboddo, *Antient Metaphysics: or, The Science of Universals*, 6 vols (1779–99; reprint, New York: Garland Publishing, 1977), IV, 26. All further references are to this edition and appear in the text. For further discussions of the Ourang-Outang by Lord Monboddo, see *Of the Origin and Progress of Language*, 2nd edn, Vol. I (Edinburgh: J. Balfour, 1774).
26. See Michel Foucault, *Madness and Civilization: A History of Insanity in the Age of Reason*, trans. Richard Howard (New York: Pantheon Books, 1965); and Edward W. Said, *Culture and Imperialism* (New York: Alfred A. Knopf, 1993).
27. See Irwin, *Mystery to a Solution*, for his discussion of the mind–body relation in the detective story, in Poe's stories and, specifically, in "The Murders in the Rue Morgue."
28. For the classic psychoanalytic reading of Poe's works, see Marie Bonaparte, *The Life and Works of Edgar Allan Poe: A Psycho-Analytic Interpretation*, trans. John Rodker (London: Imago Publications, 1949). For a discussion of Doubling in "The Murders in the Rue Morgue," see Irwin, *Mystery to a Solution*, and his discussions of the motif in this story and in detective fiction generally.
29. *Charles Darwin's Notebooks, 1836–1844: Geology, Transmutation of Species, Metaphysical Enquiries*, eds Paul H. Barrett, Peter J. Gautrey, Sandra Herbert, David Kohn, and Sydney Smith (Ithaca: Cornell University Press, 1987), pp. 263–64. I have followed the notation system used by the editors.
30. In "The Nebular Hypothesis and the Science of Progress," Simon Schaffer argues that Darwin, Robert Chambers, and John Stuart Mill, as well as Poe, were all persuaded by John Pringle Nichol's version of the nebular hypothesis (see pp. 144–45).

31. Robert Chambers, *"Vestiges of the Natural History of Creation" and Other Evolutionary Writings*, ed. James A. Secord (Chicago: University of Chicago Press, 1994), pp. 26, 360. All further references are to this edition and appear in text. For discussions of *Vestiges*, see Milton Millhauser, *Just Before Darwin: Robert Chambers and "Vestiges"* (Middletown, Conn.: Wesleyan University Press, 1959); and James A. Secord, "Behind the Veil: Robert Chambers and *Vestiges*," in *History, Humanity and Evolution*, pp. 165–94; and Secord, *Victorian Sensation: The Extraordinary Publication, Reception, and Secret Authorship of "Vestiges of the Natural History of Creation"* (Chicago: University of Chicago Press, 2000).
32. See Dwight Thomas and David K. Jackson, *The Poe Log: A Documentary Life of Edgar Allan Poe, 1809–1849* (Boston: G. K. Hall, 1987), pp. 723–24. All further references appear in the text. In "The Nebular Hypothesis and the Science of Progress" Schaffer argues that Poe heard John Pringle Nichol's New York lectures of 1847–48 (p. 145).
33. [Adam Sedgwick], review of *Vestiges of the Natural History of Creation*, [by Robert Chambers], *The Edinburgh Review*, 82 (July 1845): 25–26, 39.
34. See Irwin, *Mystery to a Solution*, on labyrinths in detective fiction. For a discussion of "The Gold-Bug," see Michael J. S. Williams, *A World of Words: Language and Displacement in the Fiction of Edgar Allan Poe* (Durham, N.C.: Duke University Press, 1988), pp. 127–40.

Chapter 2 "The Gold-Bug," Hieroglyphics, and the Historical Imagination

1. For a popular account of the decipherment of the Rosetta Stone, see Maurice Pope, *The Story of Decipherment: From Egyptian Hieroglyphs to Maya Script*, rev. edn (London: Thames and Hudson, 1999), pp. 60–84.
2. Edgar Allan Poe, "The Murders in the Rue Morgue," in *Edgar Allan Poe: Poetry and Tales*, ed. Patrick F. Quinn (New York: Library of America, 1984), p. 397. All further references to this and other Poe tales are to this edition and appear in the text.
3. See J. Hillis Miller, "Line," in *Ariadne's Thread: Story Lines* (New Haven: Yale University Press, 1992), pp. 1–27.
4. For a history of William Kidd, see Robert C. Ritchie, *Captain Kidd and the War against the Pirates* (Cambridge, Mass.: Harvard University Press, 1986).
5. See Mary P. Winsor, "Jan Swammerdam," in vol. 13 of *Dictionary of Scientific Biography*, ed. Charles Coulston Gillispie (New York: Charles Scribner's Sons, 1976), pp. 168–75. For discussions of the emerging Enlightenment concensus, see John Hedley Brooke, *Science and Religion: Some Historical Perspectives*, Cambridge History of Science, ed. George Basalla (Cambridge, Eng.: Cambridge University Press, 1991); David C. Lindberg and Ronald L. Numbers, eds, *God and Nature: Historical Essays on the Encounter between Christianity and Science* (Berkeley: University of California Press, 1986); Steven Shapin and Simon Schaffer, *Leviathan and the Air-Pump: Hobbes, Boyle, and the Experimental Life* (Princeton: Princeton University Press, 1985); Steven Shapin, *The Scientific Revolution* (Chicago: University of Chicago Press, 1996); and Stephen Toulmin, *Cosmopolis: The Hidden Agenda of Modernity* (New York: Free Press, 1990).